

Bank Customer Churn Analysis

End-to-End Data Analytics Project | Snowflake · Power BI · DAX · SQL

Project Overview

This project analyzes customer churn behavior for a bank dataset of 10,000 customers. The goal was to identify churn drivers, classify customers into behavioral segments, build a data-driven risk scoring model, and deliver a three-page interactive Power BI report for business stakeholders.

Overall churn rate: 20.37% | Balance lost to churn: €185.59M

Tech Stack

Layer	Tool
Cloud Storage	AWS S3
Data Warehouse	Snowflake
Data Preparation	SQL (Snowflake)
Visualization	Power BI Desktop
Business Logic	DAX

Data Pipeline

1. AWS S3 → Snowflake Integration

Raw CSV data was stored in an S3 bucket and loaded into Snowflake using an external stage with IAM role-based authentication.

```
sql
```

```

CREATE OR REPLACE STORAGE INTEGRATION S3_Integration
  TYPE = EXTERNAL_STAGE
  STORAGE_PROVIDER = 'S3'
  ENABLED = TRUE
  STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::819511250341:role/Snowflake_connect'
  STORAGE_ALLOWED_LOCATIONS = ('s3://project.bucket.1/');

CREATE STAGE database1.project1.stage1
  URL = 's3://project.bucket.1'
  STORAGE_INTEGRATION = s3_integration;

COPY INTO bank_data
FROM @stage1
FILES = ('Bank_Churn.csv')
FILE_FORMAT = (TYPE = csv FIELD_DELIMITER = ',' SKIP_HEADER = 1)
ON_ERROR = 'continue';

```

2. Table Schema

```

sql

CREATE TABLE IF NOT EXISTS bank_data (
  CustomerId      STRING,
  Surname         STRING,
  CreditScore     INT,
  Geography       STRING,
  Gender          STRING,
  Age             INT,
  Tenure          INT,
  Balance         FLOAT,
  NumOfProducts  INT,
  HasCrCard       INT,
  IsActiveMember INT,
  EstimatedSalary FLOAT,
  Exited          INT
);

```

Data Preparation

Bucketing Columns

Continuous variables were bucketed into categorical ranges for analysis and visualization.

```
sql
```

-- Age Buckets

```
age_bucket = CASE
    WHEN age < 25 THEN '18-24'
    WHEN age BETWEEN 25 AND 34 THEN '25-34'
    WHEN age BETWEEN 35 AND 44 THEN '35-44'
    WHEN age BETWEEN 45 AND 54 THEN '45-54'
    WHEN age BETWEEN 55 AND 64 THEN '55-64'
    ELSE '65+'
END
```

-- Tenure Buckets

```
tenure_bucket = CASE
    WHEN tenure <= 2 THEN 'New Customer'
    WHEN tenure <= 5 THEN 'Early Stage'
    WHEN tenure <= 8 THEN 'Established'
    ELSE 'Loyal'
END
```

-- Balance Buckets

```
balance_bucket = CASE
    WHEN balance = 0 THEN 'Zero Balance'
    WHEN balance < 50000 THEN 'Below 50K'
    WHEN balance < 100000 THEN '50K-100K'
    WHEN balance < 150000 THEN '100K-150K'
    WHEN balance < 200000 THEN '150K-200K'
    ELSE '200K+'
END
```

-- Credit Score Buckets (Standard FICO Ranges)

```
creditscore_bucket = CASE
    WHEN creditscore < 580 THEN 'Poor'
    WHEN creditscore < 670 THEN 'Fair'
    WHEN creditscore < 740 THEN 'Good'
    WHEN creditscore < 800 THEN 'Very Good'
    ELSE 'Excellent'
END
```

-- Salary Buckets

```
salary_bucket = CASE
    WHEN estimatedsalary < 50000 THEN 'Below 50K'
    WHEN estimatedsalary < 100000 THEN '50K-100K'
    WHEN estimatedsalary < 150000 THEN '100K-150K'
```

```
ELSE '150K+'  
END
```

Status Label Columns

sql

```
active_status = CASE WHEN isactivemember = 1 THEN 'Active' ELSE 'Inactive' EN  
exit_status   = CASE WHEN exited = 1      THEN 'Exited'  ELSE 'No'      EN  
cr_card_status = CASE WHEN hascard = 1     THEN 'Yes'     ELSE 'No'     EN
```

Risk Scoring Model

A composite risk score was built using actual churn rates per variable as weights. Each variable's component uses the **deviation from the overall churn rate (20.37%)** so all variables contribute proportionally rather than any single variable dominating the score.

The score is then normalized to a 0-100 scale.

sql

```

CREATE OR REPLACE VIEW BANK_DATA_RISK AS
WITH risk_calc AS (
    SELECT
        CUSTOMERID,
        ROUND(
            (
                (
                    CASE WHEN GEOGRAPHY = 'Germany' THEN 32.44 - 20.37 ELSE 16
                + CASE WHEN ISACTIVEMEMBER = 1 THEN 14.27 - 20.37 ELSE 26.
                + CASE
                    WHEN NUMOFPRODUCTS = 1 THEN 27.71 - 20.37
                    WHEN NUMOFPRODUCTS = 2 THEN 7.58 - 20.37
                    WHEN NUMOFPRODUCTS = 3 THEN 82.71 - 20.37
                    WHEN NUMOFPRODUCTS = 4 THEN 100 - 20.37
                    ELSE 0
                END
                + CASE WHEN GENDER = 'Female' THEN 25.07 - 20.37 ELSE 16.46 -
                + CASE
                    WHEN BALANCE > 0 AND BALANCE < 50000 THEN 34.67 - 2
                    WHEN BALANCE >= 50000 AND BALANCE < 100000 THEN 19.88
                    WHEN BALANCE >= 100000 AND BALANCE < 150000 THEN 25.77
                    WHEN BALANCE >= 150000 AND BALANCE < 200000 THEN 21.93
                    WHEN BALANCE >= 200000 THEN 55.88
                    ELSE 13.82 - 20.37
                END
                + CASE
                    WHEN AGE BETWEEN 18 AND 24 THEN 8.75 - 20.37
                    WHEN AGE BETWEEN 25 AND 34 THEN 7.76 - 20.37
                    WHEN AGE BETWEEN 35 AND 44 THEN 17.66 - 20.37
                    WHEN AGE BETWEEN 45 AND 64 THEN 49 - 20.37
                    ELSE 15.25 - 20.37
                END
                + 46.33 -- shifts minimum to 0
            ) / 213.35 -- normalizes range to 0-100
        ) * 100, 0
    ) AS RISK_SCORE,

    CASE
        WHEN isactivemember = 0 AND numofproducts > 2 THEN 'Passive Multi-
        WHEN numofproducts >= 3 THEN 'Product-Heavy'
        WHEN isactivemember = 1 AND numofproducts = 2
            AND tenure >= 4 THEN 'Committed'
        WHEN isactivemember = 1 AND numofproducts = 2

```

```

        AND tenure < 4                                THEN 'Developing'
    WHEN isactivemember = 0 AND balance >= 97000      THEN 'High-Value Sil
    WHEN isactivemember = 1 AND numofproducts = 1
        AND balance > 0                                THEN 'Growth Potenti
    WHEN isactivemember = 0 AND numofproducts <= 2
        AND balance < 97000                            THEN 'Dormant'
    ELSE 'Standard'
END AS CustomerPersona

FROM BANK_DATA
)
SELECT *,
    CASE
        WHEN RISK_SCORE >= 75 THEN 'Critical High'
        WHEN RISK_SCORE >= 50 THEN 'High'
        WHEN RISK_SCORE >= 25 THEN 'Medium'
        ELSE 'Low'
    END AS RISK_CATEGORY
FROM risk_calc;

```

Risk Score Validation

Risk Category	Total Customers	Exited	Churn Rate
Critical High	28	28	100.00%
High	327	298	91.13%
Medium	3,106	1,204	38.76%
Low	6,539	507	7.75%

Churn rates increase cleanly across all four tiers, confirming the model is predictive.

Customer Segmentation

Eight behavioral personas were defined using engagement, product, balance, and tenure signals.

Persona	Logic	Churn Rate
Committed	Active · 2 products · balance > 0 · tenure ≥ 4	5.38%
Developing	Active · 2 products · balance > 0 · tenure < 4	5.87%
Growth Potential	Active · 1 product · balance > 0	17.36%
Standard	Active · zero balance (catch-all)	26.62%
Dormant	Inactive · ≤2 products · balance < 97K	18.96%
High-Value Silent	Inactive · balance ≥ 97K	29.62%
Product-Heavy	Active · 3+ products	80.28%
Passive Multi-Product	Inactive · 3+ products	90.22%

DAX Measures & Calculated Columns

Core Measures

```
dax

Churn Rate =
DIVIDE(
    CALCULATE(COUNT(BANK_DATA[CUSTOMERID]), BANK_DATA[EXITED] = 1),
    DISTINCTCOUNT(BANK_DATA[CUSTOMERID])
)

Non Churn Rate =
DIVIDE(
    CALCULATE(COUNT(BANK_DATA[CUSTOMERID]), BANK_DATA[EXITED] = 0),
    DISTINCTCOUNT(BANK_DATA[CUSTOMERID])
)
```

Conditional Bar Color (Churn Page)

```
dax
```



```

Bar Color =
VAR ChurnRate = [Churn Rate]
RETURN
SWITCH(
    TRUE(),
    ChurnRate >= 0.5, "#FF5656", -- Red: high churn (≥50%)
    ChurnRate >= 0.25, "#062B66", -- Navy: moderate churn (25-49%)
    "#808080" -- Grey: low churn (<25%)
)

```

Value Tier Classification

```

dax

ValueTier =
VAR HighBalance = 'BANK_DATA'[Balance] > 97000
VAR HighSalary = 'BANK_DATA'[EstimatedSalary] > 100000
VAR GoodProducts = 'BANK_DATA'[NumOfProducts] >= 2
RETURN
SWITCH(TRUE(),
    HighBalance && GoodProducts, "Premium",
    HighBalance && NOT GoodProducts, "Untapped Wealth",
    NOT HighBalance && HighSalary, "Income Rich",
    NOT HighBalance && NOT HighSalary, "Standard"
)

```

Product Opportunity Classification

```

dax

ProductOpportunity =
VAR HighBalance = BANK_DATA[Balance] > 97000
VAR Products = BANK_DATA[NumOfProducts]
RETURN
SWITCH(TRUE(),
    HighBalance && Products = 1, "Cross-Sell",
    HighBalance && Products >= 2, "Well Penetrated",
    NOT HighBalance && Products >= 2, "Multi-Product",
    Products = 1, "Limited Adoption",
    "Standard"
)

```

Dashboard Structure

Page 1 — Customer Profile Overview

Demographics and distribution of the customer base. Filters: Exit Status · Activity Status · Tenure · Products Owned · Credit Card Status

Page 2 — Customer Segmentation & Opportunities

Behavioral segmentation, persona validation, risk distribution, and product opportunity analysis. Filters: Geography · Age Category · Tenure · Products Owned · Bank Balance Category · Credit Card Status · Exit Status · Activity Status

Page 3 — Customer Churn Analysis

Churn rates by key variables, exit risk validation, financial impact of churn, and retention recommendations. Filters: Customer Persona · Risk Category · Gender

Key Findings

- Germany churns at **32.44%** despite holding the highest-value customers (65.84% in 100K-150K balance band)
 - Churn peaks at ages **55-64 (49.83%)** — nearly 1 in 2 customers in this band exits
 - **2 products = safest cohort (7.58% churn)**; 3 products = 82.71%; 4 products = 100%
 - **Passive Multi-Product** customers have the highest churn of any persona at **90.22%**
 - Churned customers hold a **higher average balance (€91K)** than retained customers (€72K) — premium customer retention failure
 - **€185.59M** in balance lost to churn; €579.27M retained
 - Tenure, estimated salary, and credit score show **no meaningful variation** in churn — not valid retention signals
-

Retention Recommendations

- Prioritise inactive + high balance customers — outreach before they exit
- Germany needs a dedicated retention strategy — service quality and product fit review
- Customers with 3+ products need immediate portfolio review — over-selling is driving churn
- Cross-sell target: move 1-product customers to exactly 2 — optimal retention point

- Strengthen Early Stage onboarding — 37.61% of High Risk customers are new customers
- Investigate female churn gap (25.07% vs 16.46% male) — review product and service experience
- Do not prioritise tenure, salary, or credit score in retention efforts — no churn signal found